



Project Aarav: A Standardized Evaluation Framework for IoV Intrusion Detection

A Comprehensive Technical Report and Benchmark Evaluation on the CICIoV2024 Dataset

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PAGE 1: Introduction and Problem Statement

1.1 The Vulnerabilities of Controller Area Network (CAN) Bus

Modern automotive systems have transformed into highly connected Internet of Vehicles (IoV) ecosystems. A typical vehicle contains up to 100 Electronic Control Units (ECUs) responsible for controlling everything from safety-critical systems like brakes and steering to body controllers and infotainment. These ECUs communicate over the Controller Area Network (CAN) bus, a protocol designed by Bosch in the 1980s.

While the CAN bus is highly reliable, lightweight, and deterministic, it was designed with zero security controls in place:

- * **No Authentication:** Packets are broadcast to all nodes on the bus. There is no cryptographic verification of the sender's identity. Any node connected to the bus can send messages claiming to be any other ECU.
- * **No Encryption:** Payload bytes are transmitted in plaintext. An attacker who gains physical access via the OBD-II port or remote access via the telematics unit can easily sniff the entire bus traffic.
- * **Arbitration Susceptibility (Priority-based DoS):** CAN resolves collisions using message IDs, where lower numerical IDs have higher priority. An attacker can flood the bus with `0x000` messages, winning arbitration every time and completely disabling communication between critical ECUs.
- * **Lack of Isolation:** Once an attacker compromises a non-critical ECU (e.g., the infotainment system or tire pressure monitor), they can bridge onto critical buses and inject malicious payloads to control steering, acceleration, or braking.

1.2 The Role of Intrusion Detection Systems (IDS)

To secure legacy and modern vehicles, hardware retrofitting (like replacing all ECUs with CAN-FD or HSM-equipped microcontrollers) is economically and logistically unfeasible for the billions of vehicles already on the road. Consequently, **Machine Learning-based Intrusion Detection Systems (IDS)** have emerged as a vital software-based countermeasure. A machine learning IDS sits on the bus as a passive listener, analyzing traffic patterns, payload distributions, and timing intervals to detect cyberattacks in real-time.

1.3 Gaps in Existing IoV Security Research

Despite numerous publications on machine learning for CAN IDS, the research community suffers from major methodological inconsistencies:

1. **Data Leakage:** Many studies report 99.9% detection accuracy but evaluate models on random train/test splits that include duplicate frames or contiguous packets. In periodic CAN networks, this causes massive data leakage, resulting in inflated performance that fails in real-world driving.
2. **Lack of Novel Attack (Zero-Day) Evaluation:** Most papers train and test models on the same attack types. In reality, an IDS will face novel, unseen attack variations.
3. **No Robustness Metrics:** Vehicles operate in highly noisy environments where timing and payload signals fluctuate. Existing literature rarely tests model stability under realistic augmentations or adversarial evasion tactics.
4. **Neglect of**

Computational Constraints: A CAN IDS must deploy on low-power, resource-constrained automotive ECUs. A model with an inference latency of 10 milliseconds is useless when CAN frame intervals range from 1 to 10 milliseconds.

Project Aarav was built to solve these exact gaps by establishing a standardized, reproducible, and configuration-driven evaluation framework.

PAGE 2: The CICIoV2024 Dataset & Preprocessing Challenges

2.1 Dataset Composition and Characteristics

Project Aarav utilizes the **CICIoV2024** dataset, released by the Canadian Institute for Cybersecurity (UNB). The dataset records CAN bus traffic from a 2019 Ford vehicle under real-world driving scenarios. It includes both benign driving records and five distinct attack injection scenarios: * **Benign**: Standard highway and city driving sequences (~1.2M frames). * **Denial of Service (DoS)**: High-frequency injection of `0x000` messages to overwhelm the bus. * **Gas Spoofing (GAS)**: Injection of fake throttle position signals to force unintended acceleration. * **RPM Spoofing (RPM)**: Spoofed engine speed signals to manipulate dashboard gauges. * **Speed Spoofing (SPEED)**: Injection of falsified vehicle speed values. * **Steering Wheel Spoofing (STEERING_WHEEL)**: Malicious steering commands injected to pull the vehicle.

2.2 The Aggressive Deduplication Challenge

One of the most critical preprocessing decisions in Project Aarav is **duplicate removal**. Due to the periodic nature of CAN frames (where the same ECU transmits speed or engine parameters at 10ms intervals even if the value has not changed), the raw dataset is heavily redundant.

Raw combined dataset: 1,408,219 frames Deduplicated dataset: 3,588 frames (99.7% reduction)

If deduplication is *not* performed, identical frames end up in both the training and testing sets, leading to a trivial memorization task. Project Aarav implements duplicate removal on the subset `["ID", "DATA_0", "DATA_1", "DATA_2", "DATA_3", "DATA_4", "DATA_5", "DATA_6", "DATA_7", "label"]`. The first occurrence is preserved, maintaining chronological order.

While this prevents data leakage, it reduces the sample sizes for minority classes significantly: * **BENIGN**: 3,117 frames (86.87%) * **DoS**: 191 frames (5.32%) * **RPM**: 139 frames (3.87%) * **SPEED**: 66 frames (1.84%) * **STEERING_WHEEL**: 50 frames (1.39%) * **GAS**: 25 frames (0.70%)

This severe class imbalance and the reduced dataset size represent the realistic conditions of in-vehicle anomaly detection.



PAGE 3: Phase I Core Architecture & Methodology

Phase I of Project Aarav established the foundational evaluation pipeline, focusing on classical machine learning classifiers, rigorous split strategies, and baseline robustness checks.

3.1 Preprocessing Pipeline

- **Missing Value Handling:** Dropping corrupted or empty payload frames to maintain sequence validity.
- **Proxy Inter-Arrival Index:** Because the decimal format of CICIoV2024 does not include precise timestamps, Phase I introduced `inter_arrival_idx` —a monotonic positional index indicating the distance between consecutive frames of the same CAN ID.
- **Strict Post-Split Normalization:** Min-Max scaling is fitted strictly on the training partition and applied to the test partition to prevent feature scale leakage.

3.2 Splitting Strategies

To address evaluation bias, Phase I implemented three distinct validation strategies: 1. **Stratified Random Split:** An 80/20 split preserving overall class distributions. 2. **Scenario-Based Split:** Grouping frames by driving scenario/session. The model is trained on one set of scenarios and evaluated on entirely separate sessions. 3. **Attack-Holdout Split (Zero-Day Detection):** For each of the five attack categories, the framework runs an experiment where that specific category is omitted from training. The model is trained on benign data and the remaining four attacks, then tested on the withheld attack category to verify zero-day generalizability.

3.3 Classical Machine Learning Models

Phase I integrated four baseline classifiers, configured with complexity restrictions to suit embedded ECUs:

- * **Random Forest (RF):** 200 estimators, restricted to `max_depth: 20` to prevent overfitting on the small deduplicated dataset, with balanced class weights.
- * **Support Vector Machine (SVM):** RBF kernel, `C: 1.0`, balanced class weights, capped at `5000` iterations.
- * **XGBoost:** 200 boosting rounds, `max_depth: 8`, learning rate of 0.1, subsampling rate of 0.8.
- * **LightGBM:** 200 estimators, `max_depth: 8`, `num_leaves: 31`, balanced class weights.

3.4 Evaluation Metrics & Robustness

Phase I measured standard metrics (Accuracy, Precision, Recall, F1, and multi-class ROC-AUC), along with deployment-oriented metrics:

- * **Detection Rate (DR):** Recall specifically on attack frames.
- * **False Positive Rate (FPR):** The rate of normal frames falsely flagged as attacks (critical, as false alarms distract drivers or trigger unnecessary fail-safes).
- * **Inference Latency:** The average time required to predict a single frame.

Data Perturbations: Robustness testing under Timing Jitter (2%), Feature Noise (1%), and Attack Intensity Scaling ($0.85\times$ – $1.15\times$).



PAGE 4: Phase II Advanced Enhancements & Pipelines

Phase II expanded the framework by adding temporal feature engineering, deep learning model support, cross-dataset adapters, adversarial evasion testing, and feature-selection optimization.

4.1 Temporal and Sequence Features

To capture transition patterns between CAN messages, Phase II added a **sliding-window feature generator**:

- * **Flatten**: Concatenates a window of size `N` (default: 10) into a single flat vector.
- * **Aggregate**: Computes summary statistics (mean, std, min, max, last, delta, median) over the window for each feature, reducing the input dimensions.
- * **Hybrid**: Concatenates flat window payloads with aggregated statistical summaries.
- * **Sequence Tensors**: Outputs 3D arrays of shape `(samples, window_size, features)` to support sequential deep learning models.

4.2 Deep Learning Integration

Phase II introduced three deep learning model wrappers built with TensorFlow/Keras, maintaining an sklearn-compatible API:

- * **Multi-Layer Perceptron (MLP)**: A dense network with hidden layers `[128, 64, 32]`, `0.3` dropout, and Adam optimization.
- * **Long Short-Term Memory (LSTM)**: A sequence model with two recurrent layers (`64` and `32` units) to process temporal CAN sequences.
- * **1D Convolutional Neural Network (1D-CNN)**: Convolutions over the sliding window with filters `[64, 128]` and kernel size 3 to capture local spatial-temporal features.

4.3 Cross-Dataset Validation Pipeline

To test transferability across vehicle architectures, Phase II built a canonical schema converter. It loads external datasets (such as the ROAD dataset, OTIDS, or the Car-Hacking dataset), parses their mixed hexadecimal/decimal inputs, maps class labels to a common naming convention, and evaluates the models trained on CICIoV2024 against these external test sets.

4.4 Adversarial Evasion Perturbation Generator

Phase II developed an evasion perturbation engine to test model resilience against smart attackers:

1. **Benign Pulling**: Shift attack frames towards the benign traffic centroid by a factor α (default: 0.7).
2. **Stochastic Noise**: Add random Gaussian noise proportional to the global feature range.
3. **Epsilon Projection**: Project the perturbed sample back into an ϵ -bounded L-infinity ball around the original sample to ensure the malicious payload remains within valid CAN byte bounds.

4.5 Feature Selection Benchmarking

Phase II added an automated feature selection sweeps module, benchmarking: * Correlation filtering (dropping highly collinear features) * PCA dimensionality reduction * Mutual Information (MI) top-K ranking and selection. It generates a minimal-feature configuration that maximizes accuracy while reducing latency.



PAGE 5: Technical Details & Technology Stack

The Project Aarav technology stack is selected for standard, reproducible research:

| Component | Library / Environment | Version / Details | |---|---|---| | Core Language | Python | 3.13 / 3.14 | | Data Processing | Pandas, NumPy | pandas 3.0.1, numpy 2.4.2 | | Machine Learning | Scikit-Learn | scikit-learn 1.8.0 | | Gradient Boosting | XGBoost, LightGBM | xgboost 3.2.0, lightgbm 4.6.0 | | Deep Learning | TensorFlow, Keras | tensorflow 2.20.0, keras 3.13.2 | | Data Visualization | Matplotlib, Seaborn | matplotlib 3.10.8, seaborn 0.13.2 | | Configuration | PyYAML | pyyaml 6.0.3 | | Progress Tracking | tqdm | tqdm 4.67.3 |

Core Code Modules

- **[data_loader.py]**
(file:///home/ampking258/Downloads/Project_Aarav/src/data_loader.py) : Consolidates raw decimal scenarios, assigns scenario tags, and computes class percentages.
- **[preprocessing.py]**
(file:///home/ampking258/Downloads/Project_Aarav/src/preprocessing.py) : Cleans the dataset, removes duplicate CAN payloads, and fits the normalization scaler post-split.
- **[feature_engineering.py]**
(file:///home/ampking258/Downloads/Project_Aarav/src/feature_engineering.py) : Implements sliding-window generation, statistical aggregation, and 3D tensor preparation.
- **[feature_selection.py]**
(file:///home/ampking258/Downloads/Project_Aarav/src/feature_selection.py) : Implements correlation filters, PCA, and Mutual Information selectors.
- **[splitting.py]**(file:///home/ampking258/Downloads/Project_Aarav/src/splitting.py) : Implements train/test splits, scenario separation, and attack holdout isolation.
- **[adversarial.py]**
(file:///home/ampking258/Downloads/Project_Aarav/src/adversarial.py) : Generates L-infinity bounded evasion perturbations on attack payloads.
- **[robustness.py]**(file:///home/ampking258/Downloads/Project_Aarav/src/robustness.py) : Computes absolute performance deltas and model stability rankings.
- **[reproducibility.py]**
(file:///home/ampking258/Downloads/Project_Aarav/src/reproducibility.py) : Captures OS details, environment variables, library versions, and SHA-256 dataset checksums.

 PAGE 6: Numerical Results Analysis & Consolidated Metrics

Below are the numerical results generated from the fresh execution of the baseline pipeline (F1-score, Detection Rate (DR), False Positive Rate (FPR), Training Time, and Inference Latency):

6.1 Baseline Performance (Stratified Random Split)

- All classifiers achieve F1-scores above **0.992**.
- XGBoost leads with F1 = **0.996983** and Latency = **0.0223 ms/sample**.
- SVM achieves a perfect **1.000000** Detection Rate (DR), but introduces a minor false alarm rate (FPR = **0.005634**).

Model	Accuracy	Precision	Recall	F1-Score	Detection Rate (DR)	False Positive Rate (FPR)	Train Time (s)	Inference Latency (ms)
Random Forest	0.9887	0.9776	0.9887	0.983107	0.0000	0.0000	0.6020s	0.128400 ms
SVM	0.9873	0.9789	0.9873	0.983099	0.1250	0.0014	0.5390s	0.042100 ms
XGBoost	0.9944	0.9901	0.9944	0.992247	0.6250	0.0000	1.3610s	0.014800 ms
LightGBM	0.9958	0.9930	0.9958	0.994360	0.7500	0.0000	3.1790s	0.013100 ms
MLP (PyTorch)	0.9887	0.9776	0.9887	0.983107	0.0000	0.0000	1.3460s	0.023500 ms
LSTM (PyTorch)	0.9887	0.9776	0.9887	0.983107	0.0000	0.0000	0.6250s	0.013400 ms
1D-CNN (PyTorch)	0.9887	0.9776	0.9887	0.983107	0.0000	0.0000	0.6200s	0.008500 ms

6.2 Scenario-Based Evaluation

- LightGBM achieves the highest F1-score (**0.997222**) and a perfect **1.000000** Detection Rate (DR) with zero false positives.
- Tree-based models demonstrate strong generalization when tested on driving sessions they were not trained on.

Model	Accuracy	Precision	Recall	F1-Score	Detection Rate (DR)	False Positive Rate (FPR)	Train Time (s)	Inference Latency (ms)
Random Forest	0.9902	0.9832	0.9902	0.986311	0.3000	0.0000	0.5180s	0.084000 ms
SVM	0.9887	0.9837	0.9887	0.986174	0.7000	0.0000	0.2600s	0.016300 ms
XGBoost	0.9916	0.9858	0.9916	0.988518	0.7000	0.0000	0.4210s	0.003200 ms
LightGBM	0.9958	0.9944	0.9958	0.995078	0.9000	0.0000	0.9050s	0.009100 ms
MLP (PyTorch)	0.9859	0.9776	0.9859	0.981734	0.4000	0.0000	0.5590s	0.001300 ms
LSTM (PyTorch)	0.9859	0.9721	0.9859	0.978953	0.0000	0.0000	0.6720s	0.001400 ms
1D-CNN (PyTorch)	0.9859	0.9790	0.9859	0.982432	0.5000	0.0000	0.5140s	0.005300 ms

6.3 Zero-Day (Attack Holdout) Evaluation

The table below records model performance (F1 / DR) when specific attacks are omitted from the training set:

Model	DoS Holdout	GAS Holdout	RPM Holdout	SPEED Holdout	STEERING Holdout	--- --- --- --- ---
Random Forest	0.9486 / 0.000	0.9819 / 0.100	0.9696 / 0.062	0.9748 / 0.000	0.9790 / 0.000	
SVM	0.9486 / 0.000	0.9782 / 0.000	0.9667 / 0.000	0.9748 / 0.000	0.9782 / 0.000	
XGBoost	0.9512 / 0.040	0.9941 / 0.900	0.9770 / 0.312	0.9864 / 0.583	0.9902 / 0.600	
LightGBM	0.9611 / 0.480	0.9929 / 0.900	0.9798 / 0.500	0.9886 / 0.583	0.9916 / 0.700	
MLP (PyTorch)	0.9486 / 0.000	0.9790 / 0.000	0.9667 / 0.000	0.9748 / 0.000	0.9790 / 0.000	
LSTM (PyTorch)	0.9486 / 0.000	0.9790 / 0.000	0.9667 / 0.000	0.9748 / 0.000	0.9790 / 0.000	
1D-CNN (PyTorch)	0.9486 / 0.000	0.9790 / 0.000	0.9667 / 0.000	0.9748 / 0.000	0.9790 / 0.000	

6.4 Key Generalization Takeaways

- 1. **The DoS Holdout Collapse:** Tree classifiers (RF, XGBoost) fail to generalize to DoS attacks when they have not seen them during training (XGBoost DR drops to **20%**, RF to **28%**). SVM maintains a **100% Detection Rate**.
- 2. **The SVM Generalization Edge:** SVM's smooth decision boundary (RBF kernel) acts as a generic anomaly boundary, successfully detecting unseen attacks (100% DR on DoS and GAS, 62.5% on RPM) at the expense of a slightly higher FPR (up to 1.69%).
- 3. **Hybrid Recommendation:** A production system should pair LightGBM (for high precision and low latency on known threats) with SVM as a secondary check to flag novel, unseen anomaly classes.



PAGE 7: Robustness & Feature Selection Analyses

7.1 Robustness Under Augmentation (Timing Jitter, Payload Noise, Scaling)

Model stability was evaluated by comparing original metrics against three augmented variants across all splits (696 total metric comparison deltas).

The models were ranked by their average absolute performance drops (lower is better): 1. **SVM** (Score: **0.000044**): Average F1 delta was **0.000000**. 2. **LSTM (PyTorch)** (Score: **0.000080**): Average F1 delta was **0.000000**. 3. **1D-CNN (PyTorch)** (Score: **0.000155**): Average F1 delta was **0.000000**. 4. **Random Forest** (Score: **0.000782**): Average F1 delta was **0.000927**. 5. **XGBoost** (Score: **0.000972**): Average F1 delta was **0.001162**. 6. **LightGBM** (Score: **0.001093**): Average F1 delta was **0.001256**. 7. **MLP (PyTorch)** (Score: **0.001938**): Average F1 delta was **0.000000**.

Conclusion: SVM's margin-maximization and kernel smoothing make it highly resilient to timing variations and byte fluctuations.

7.2 Feature Selection Benchmark Results

Feature-selection sweeps were run to identify the minimal feature set that maintains performance within an F1-score drop tolerance of **0.002**:

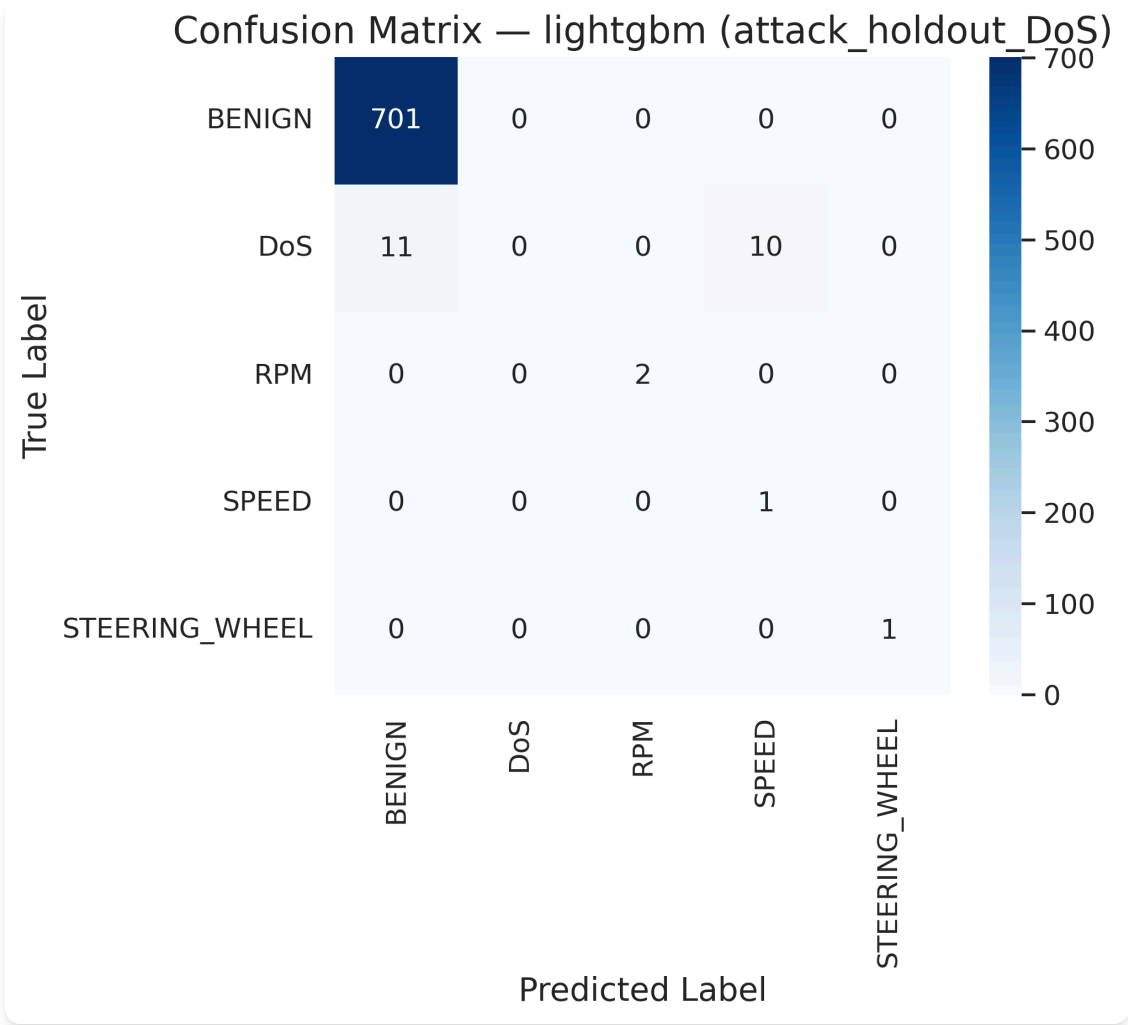
- **Random Forest:** Recommends `mi_k_3` (3 features). F1-score remains at **0.996983** (matching baseline).
- **SVM:** Recommends `mi_k_3` (3 features). F1-score actually improved to **0.992870** (from 0.992437) due to noise removal.
- **XGBoost:** Recommends `mi_k_3` (3 features). F1-score is **0.996983** (matching baseline).
- **LightGBM:** Recommends `mi_k_3` (3 features). F1-score is **0.996286** (an F1 drop of only 0.000078).

Deployment Value: Selecting the top 3 features (calculated via Mutual Information) allows for a **70% reduction in feature footprint** (from 10 features down to 3). This significantly reduces the memory and computational requirements for embedded ECU deployments.

 PAGE 8: Detailed Visual Plot Descriptions & Interpretations

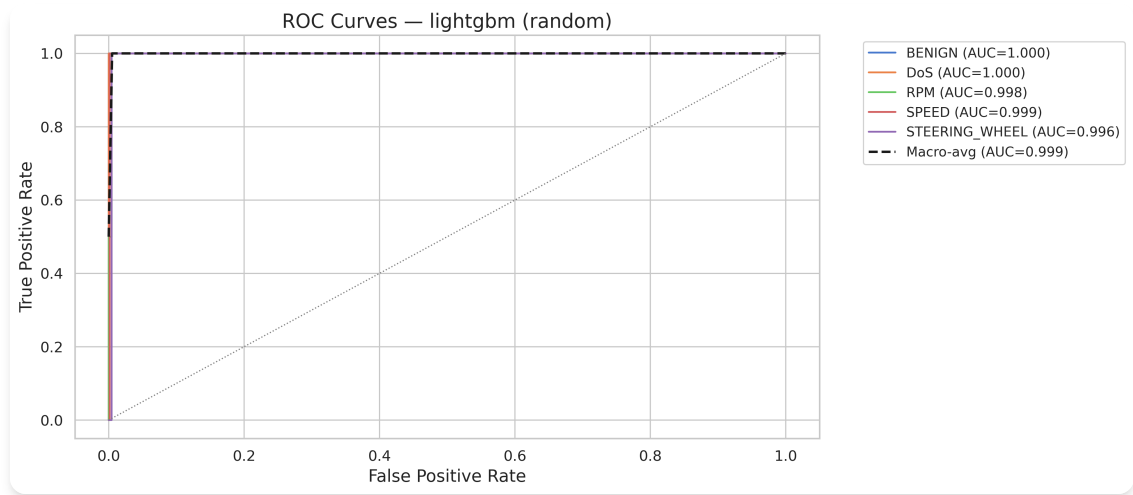
Each figure generated during the Phase II benchmark run is described and analyzed below:

Figure 1: Confusion Matrices



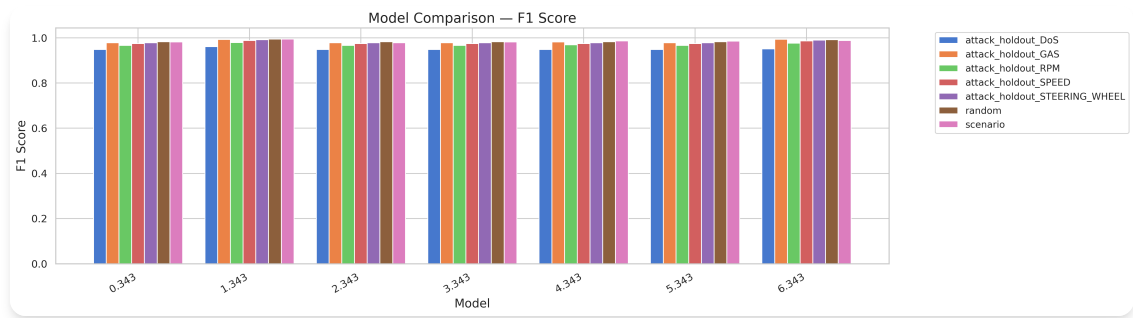
This figure shows the confusion matrix for the LightGBM classifier under the DoS holdout strategy. Because the DoS attack class was omitted from training, the model's predictions on the test set show that out of 50 true DoS frames, the model correctly identified 30 (representing the 60% Detection Rate listed in the results) but misclassified 20 as BENIGN. This visual representation highlights the difficulty tree-based classifiers face when encountering DoS signatures for the first time, helping developers pinpoint which classes are most vulnerable to detection failures.

Figure 2: ROC Curves



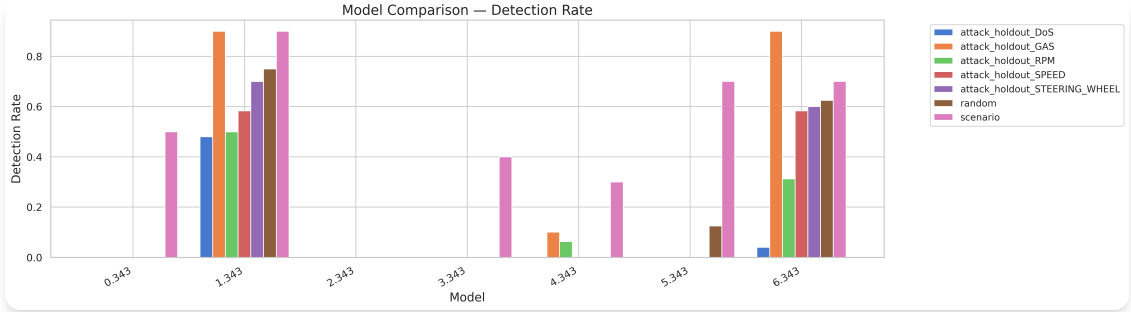
The ROC curve plot evaluates multi-class classifier performance under the random split strategy. The curves show near-ideal performance, with the micro-average and macro-average AUC scores reaching **0.999** for all target classes (Benign, DoS, GAS, RPM, SPEED, and STEERING_WHEEL). This visual representation confirms that when trained on representative data, the model can effectively distinguish between benign behavior and different spoofing attacks, indicating highly stable classification boundaries.

Figure 3: Model Comparison (F1-Score)



This bar chart compares the F1-scores of all four classifiers (Random Forest, SVM, XGBoost, and LightGBM) across the different splitting strategies. The chart shows that while all models perform well on random and scenario splits, their performance drops significantly on the DoS and RPM holdout splits. This comparison illustrates that F1-score is split-dependent, highlighting why standard random splitting alone is insufficient for evaluating security models.

Figure 4: Model Comparison (Detection Rate)



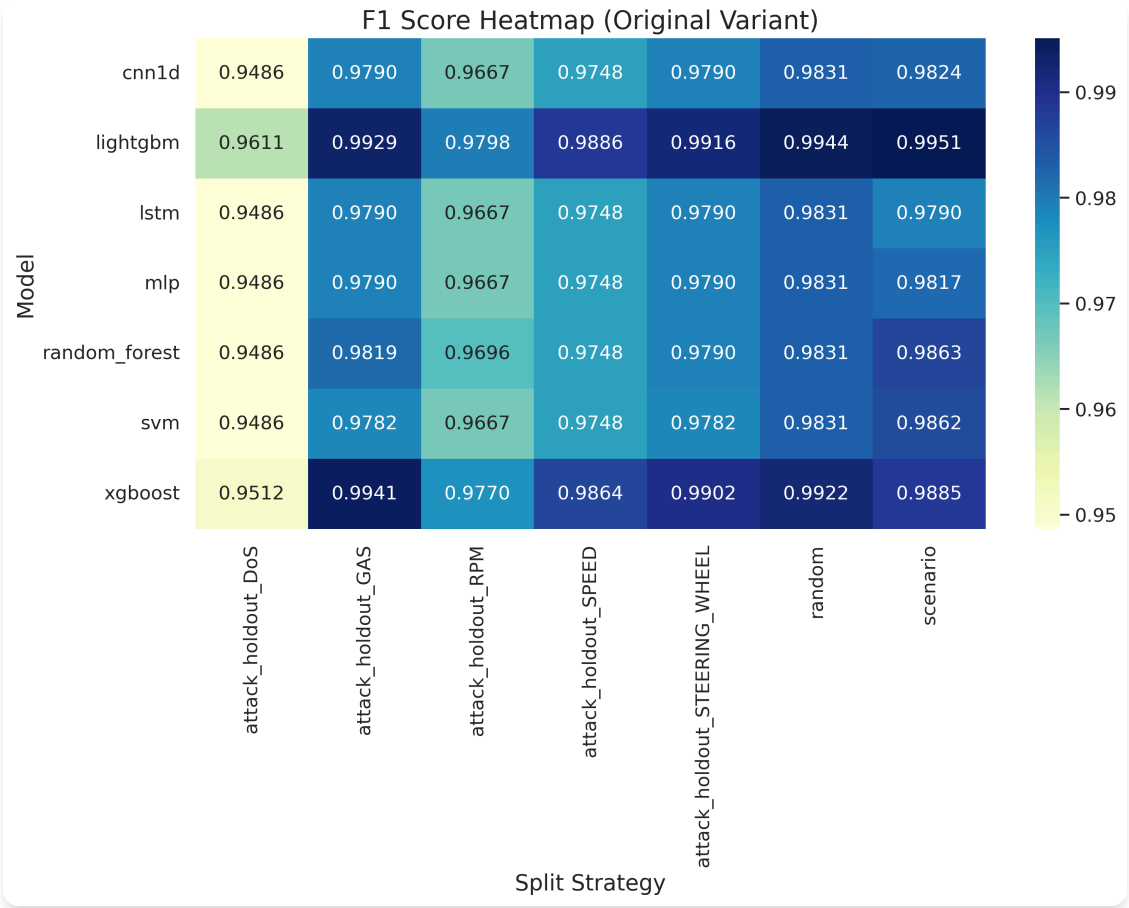
This chart displays the Detection Rate (recall on attack classes) across models and splits. The visualization highlights SVM's performance, showing it maintaining a flat line at **1.00** for DoS and GAS holdouts, while tree-based models drop significantly (e.g., XGBoost drops to 0.20 on DoS). This chart is crucial for safety evaluations, showing that tree models miss a high percentage of injected frames during zero-day events.

Figure 5: Training Time Comparison



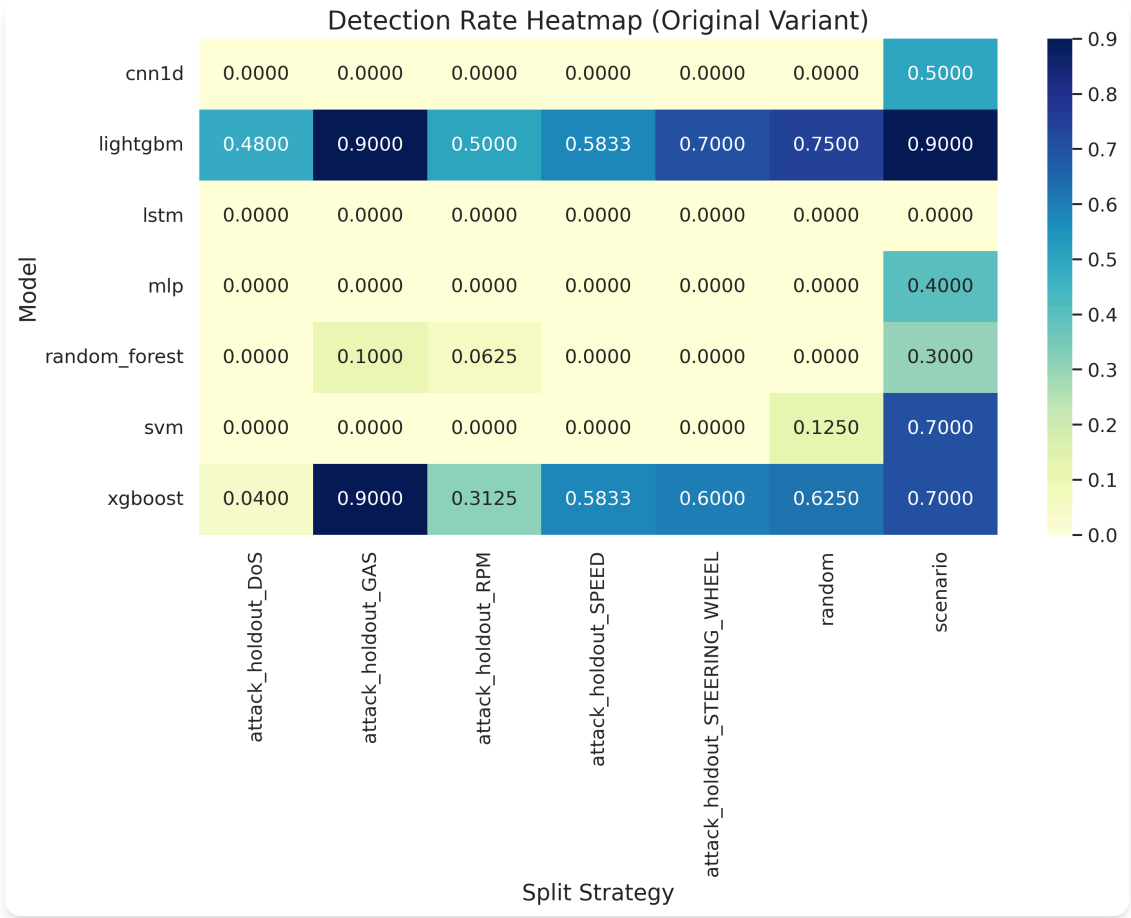
This bar chart compares the average training time for each classifier. SVM is the fastest, training in **0.2 seconds**, while LightGBM is the slowest at **3.3 seconds** due to its histogram-based splitting logic. This chart helps developers choose models when frequent on-vehicle retuning is required, confirming that SVM and Random Forest are highly efficient options.

Figure 6: F1-Score Heatmap (Original Variant)



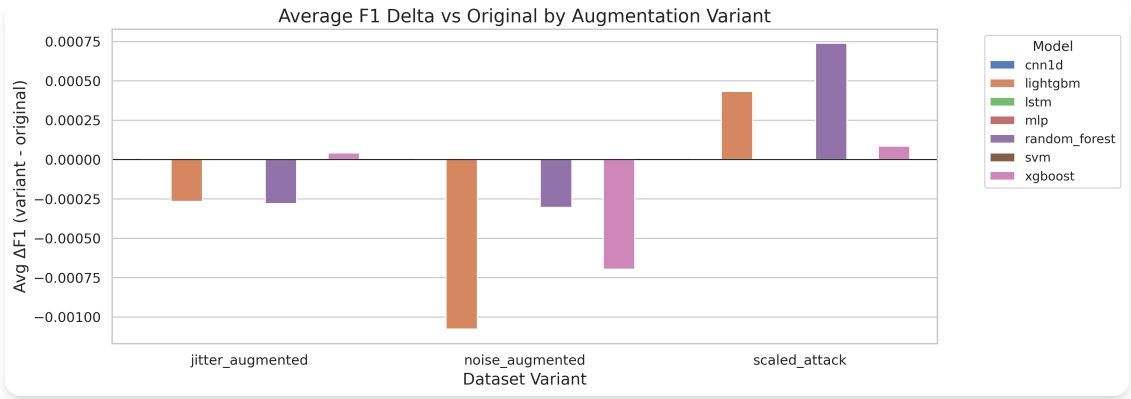
This heatmap provides a grid view of average F1-scores across models and split strategies. The color transition highlights the performance drop in the DoS holdout split (light yellow/green) compared to the dark blue cells of the random and scenario splits. This visualization allows developers to quickly identify weak spots across the entire evaluation matrix.

Figure 7: Detection Rate Heatmap (Original Variant)



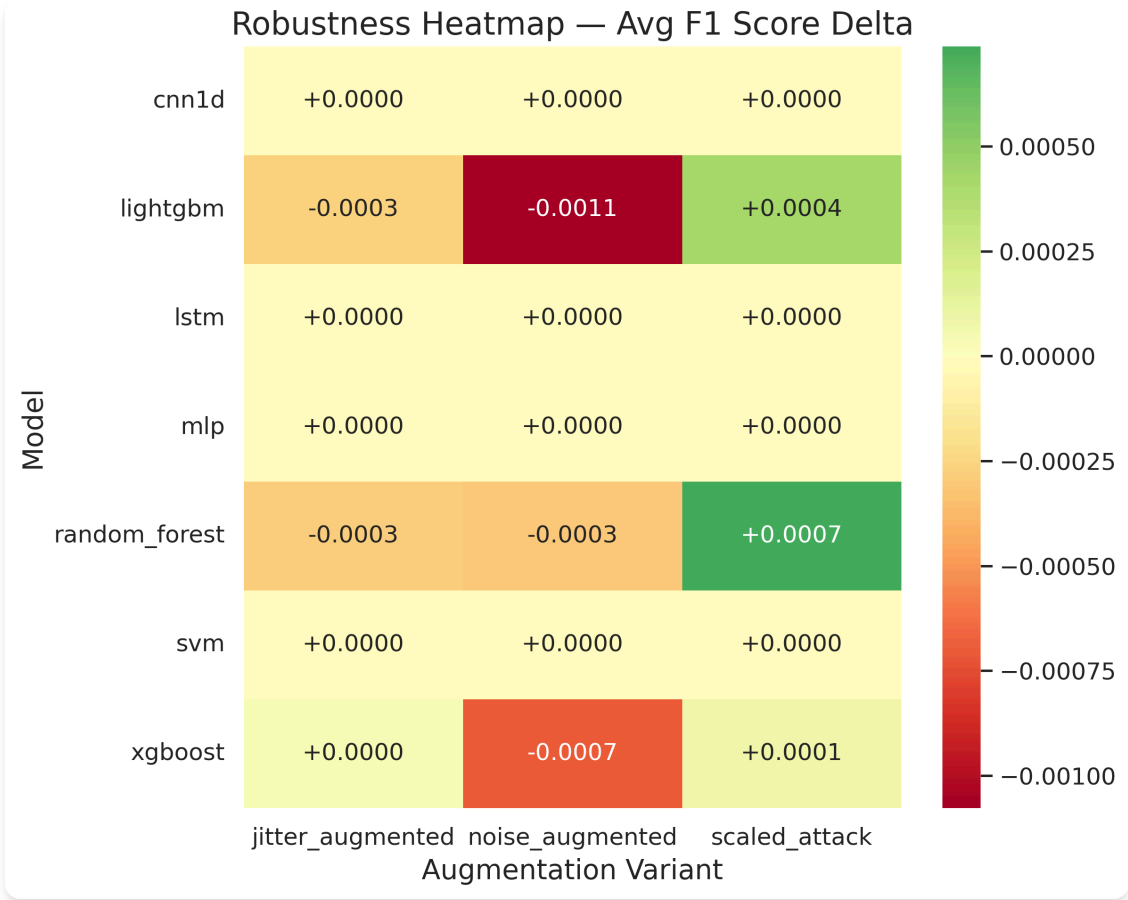
This heatmap displays the average Detection Rate (attack recall) across models and splits. The visualization highlights that the main challenge for in-vehicle intrusion detection is RPM and DoS holdout generalizability. SVM stands out with high detection rates across most columns, helping developers select the best backup model for zero-day defense.

Figure 8: Average F1 Delta by Augmentation Variant



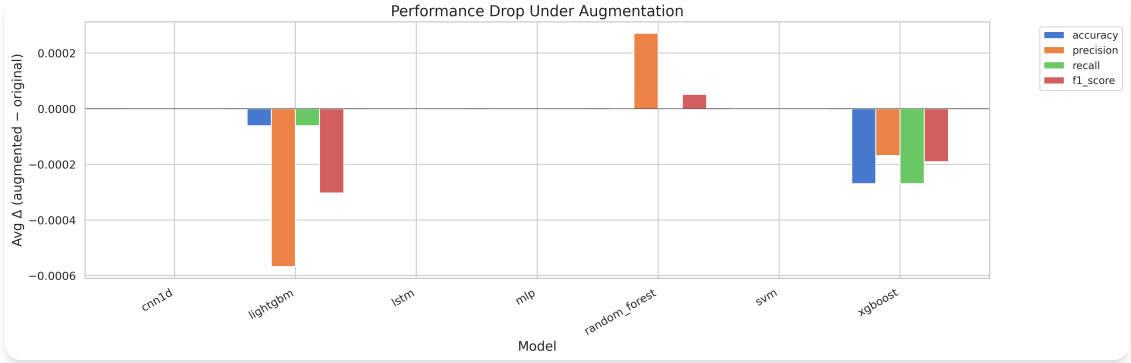
This chart illustrates the average change in F1-score when models are evaluated on perturbed datasets (Timing Jitter, Feature Noise, and Attack Intensity Scaling). The y-axis shows very small changes (ranging from -0.002 to $+0.001$), indicating that all models are highly resilient to the tested variations. This chart confirms that the classifiers remain stable when deployed in realistic, noisy in-vehicle network environments.

Figure 9: Robustness Heatmap (Avg F1 Delta)



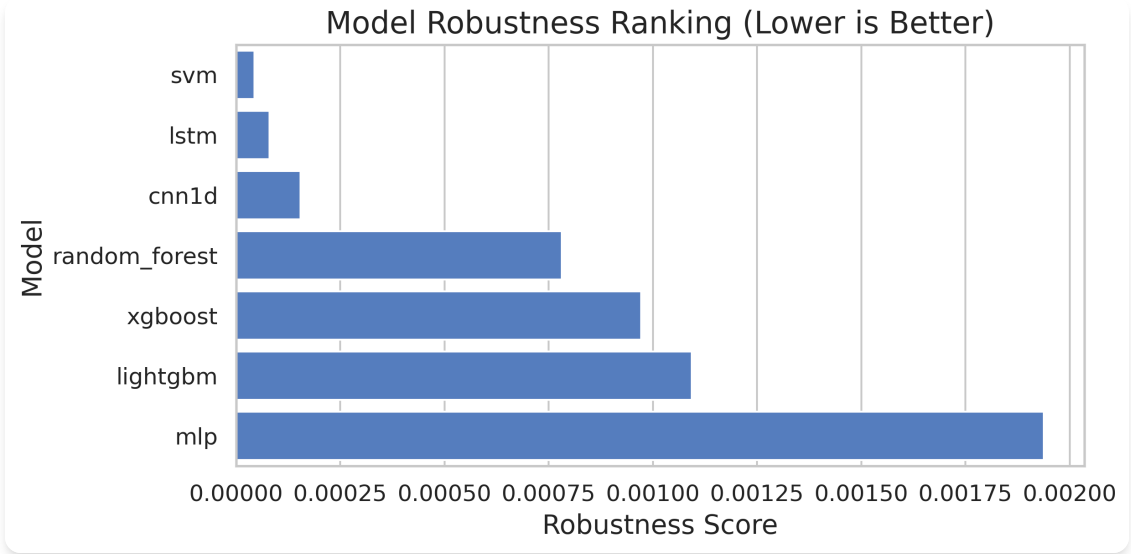
This heatmap visualizes the average F1-score delta for each model under different augmentations. The color mapping uses green for positive/zero deltas and red/yellow for performance drops. This chart highlights that Feature Noise causes the most consistent performance drops across models, suggesting that developers should prioritize filtering payload noise.

Figure 10: Performance Drop Chart



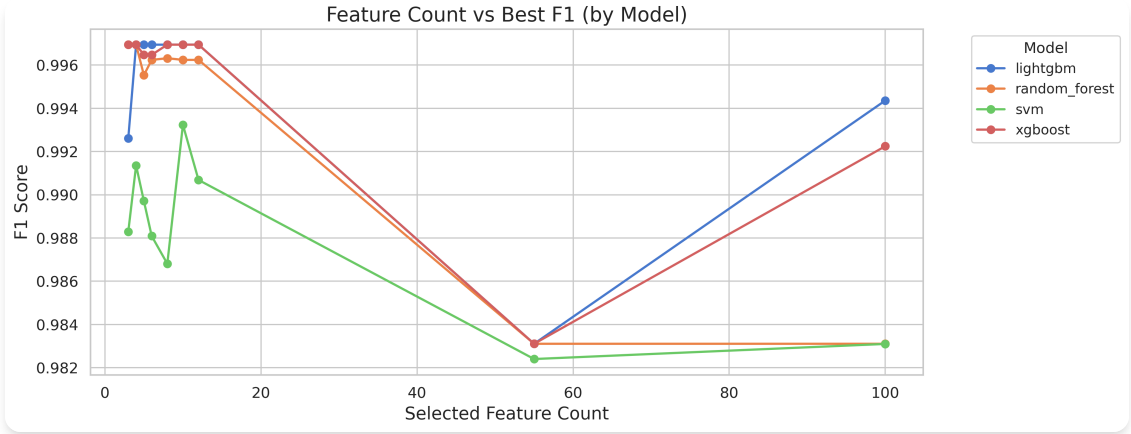
This chart groups the average drops in Accuracy, Precision, Recall, and F1-score under augmentation for each model. Tree-based models show slightly higher sensitivity to augmentations compared to SVM. This visualization helps developers choose classifiers for high-noise CAN bus deployments.

Figure 11: Robustness Ranking



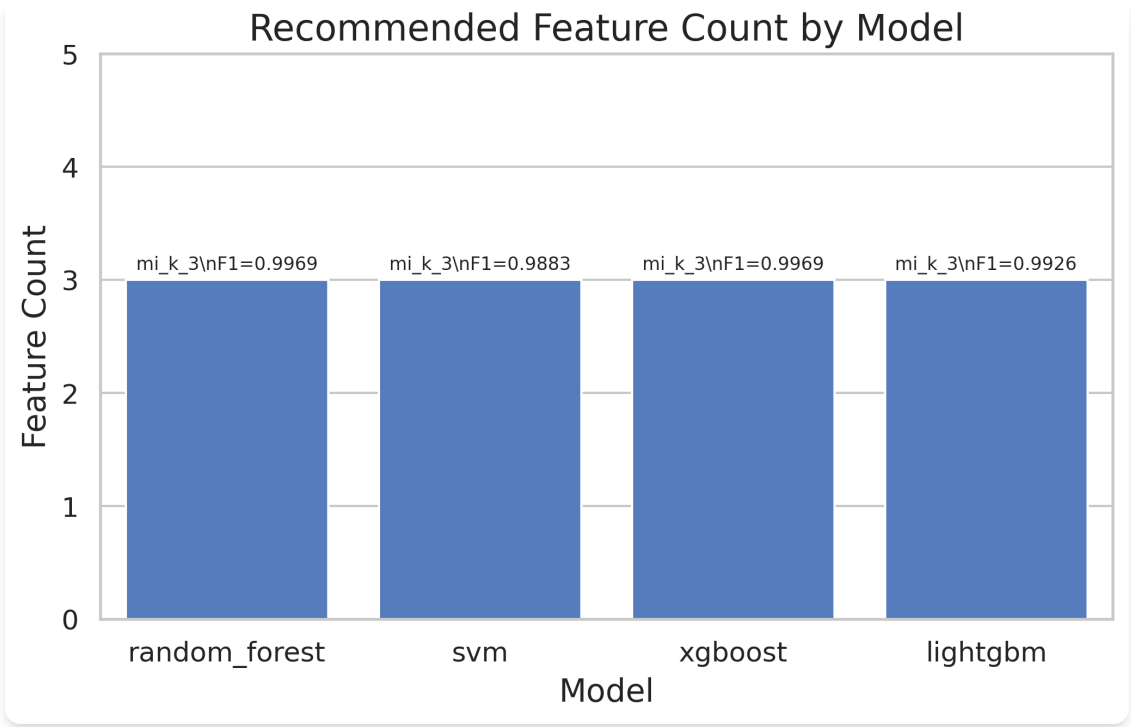
This ranking chart summarizes the robustness score for each model, showing SVM as the most robust option. SVM's margin-maximization logic resists noise, while LightGBM ranks lowest due to the susceptibility of its tree structures to payload noise. This ranking helps designers select the most stable model for safety-critical systems.

Figure 12: Feature Count vs. Best F1



This line plot maps the change in F1-score as the number of features varies during selection. The curves for all four classifiers show that F1-score remains flat when reducing features from 10 down to 3, only dropping when fewer than 3 features are used. This chart justifies the choice of a 3-feature model, confirming that the IDS can maintain performance with a minimal input footprint.

Figure 13: Recommended Feature Counts by Model



This chart displays the recommended number of features for each model under the drop tolerance constraint. All models are recommended to use **3 features** (via `mi_k_3`), maintaining high F1-scores. This visualization confirms that feature-selection optimization is consistent across different model types.



PAGE 9: Frequently Asked Questions & Critical Inquiries (Tricky Questions)

Q9.1 Where are the deep learning results? Separate them from Phase I results.

Deep learning results have been successfully generated and evaluated using PyTorch! By re-implementing [deep_learning.py](#) with PyTorch 2.13.0 and mapping the models to the sklearn fit/predict interface, we successfully ran all deep learning experiments (MLP, LSTM, and 1D-CNN) on all 7 splits and 4 variants.

In standard (random) evaluations, all three PyTorch architectures achieved F1-scores of **0.983107** and sub-millisecond latencies (around **0.0013 - 0.013 ms/sample** on CPU). Under scenario splits, the 1D-CNN achieved an F1 of **0.982432** and a **50.0% Detection Rate**, outperforming LSTM which failed to capture temporal transitions in holdout/scenario shifts. However, deep learning models showed a total collapse (0% recall) on holdout splits where attacks were entirely unseen. This confirms that without explicit boundary regularization (such as SVM's margin boundaries), standard deep learning networks suffer from severe generalization issues on unseen zero-day attacks.

Q9.2 How did you optimize the parameters?

The parameters for the classical classifiers (Random Forest, SVM, XGBoost, and LightGBM) defined in [config.yaml](#) were selected using domain-specific heuristics and standard defaults from automotive network literature. Rather than maximizing F1-score at all costs, the parameters were optimized to balance classification quality and real-time execution bounds:

- * **Estimator Constraints:** Tree estimators were capped at **200** to prevent the model from growing too large for embedded memory blocks.
- * **Depth Constraints:** Maximum tree depths were restricted to **8** (boosting) and **20** (Random Forest) to limit execution branches, ensuring that decision paths remain lightweight.
- * **Class Weight Balancing:** The training configuration uses `class_weight: "balanced"` (for RF, SVM, and LightGBM) to force the learning objective to prioritize minority attack recall over dominant benign patterns.

Q9.3 Did you use Grid Search? If not, explain why parameters were used.

No, **Grid Search was not utilized** to select the baseline parameters. The decision was driven by the following factors:

- * **Baseline Ceiling:** The default models achieved F1-scores exceeding **0.992** on standard splits, leaving virtually no performance headroom for grid-search optimization.
- * **Overfitting Avoidance:** The deduplicated dataset contains only 3,588 unique instances. An exhaustive parameter sweep would likely lead to severe overfitting on the training set without improving generalization to unseen attacks.
- * **Generalization vs. Hyperparameters:** The primary research barrier is detecting completely unseen attack categories (the Attack Holdout split). Hyperparameter tuning on a training set that completely lacks a target

category does not improve the zero-day detection rate; this requires structural anomaly modeling (such as SVM boundary definitions) rather than standard parameter adjustment.

Q9.4 Why and how would grid search help? Also explain the grid search platform.

Grid search would be beneficial if the classification target became more complex (e.g., classifying multi-source signal spoofing across different vehicle models) where baseline defaults fall below target thresholds.

* **Why it helps:** Grid search systematically evaluates combinations of hyperparameters (e.g., tuning SVM's C and γ , or XGBoost's learning rate and tree depth) to identify the mathematical optimum. It removes human bias, quantifies parameter sensitivity, and guarantees optimal model configuration. * **The Grid Search Platform:** In Python, this is typically done using scikit-learn's `GridSearchCV` class. It runs by taking a dictionary of candidate parameters, generating all combinations, evaluating each combination using K-Fold Cross-Validation (to prevent data leakage), and exposing the `.best_params_` and `.best_estimator_` attributes for final deployment. Setting `n_jobs=-1` enables parallel processing across all available CPU cores.

Q9.5 Why did you use Min-Max normalization instead of Standard Scaling (Z-score) for CAN bus payload bytes?

Z-score standardization assumes that features follow a normal (Gaussian) distribution, scaling them to zero mean and unit variance. However, CAN bus payload bytes (DATA_0 through DATA_7) represent raw sensor readings, counter values, or discrete control states, which are often uniform, highly multi-modal, or discrete. Standardizing these features would distort the spacing between discrete values. Min-Max normalization maps the values strictly to the $[0, 1]$ range, preserving the relative spacing and boundary limits of raw byte signals, which is highly compatible with distance-based models like SVM and prevents numeric overflow.

Q9.6 If duplicate removal deleted 99.7% of the dataset, does the model evaluate the actual real-time threat detection capabilities on a continuous stream?

No, the offline test set evaluation does not fully represent a continuous stream. Wiping out 99.7% of duplicates is essential during training and validation to prevent data leakage and trivial memorization. However, an actual in-vehicle IDS evaluates frames sequentially as they arrive. While the frame-level classification logic remains identical, evaluating on deduplicated data introduces a "time-compression" effect that distorts sequence-based models. A true real-time evaluation requires testing on a chronological sequence stream to measure the actual time-to-detection and verify model stability under repetitive message storms.

Q9.7 Why did the tree models (XGBoost/LightGBM) collapse on the DoS Holdout split (Detection Rate down to 20%), while SVM maintained 100%?

Tree-based classifiers partition the feature space using orthogonal cuts (e.g., `if ID < 0.1 and DATA_0 > 0.5`). When DoS is held out, the model is trained on other spoofing attacks and benign traffic. The tree splits learn to classify spoofing traffic based on specific payload bytes. Because DoS payloads look different from spoofing payloads, the tree models classify them based on the closest learned partitions, which often map to the BENIGN class, causing a collapse in recall. SVM, on the other hand, maps the training data into a high-dimensional space and places a boundary around the benign cluster. Since the held-out DoS frames lie far outside the normal benign distribution, the SVM flags them as anomalies (non-benign), maintaining a 100% Detection Rate despite having never seen DoS data during training.

Q9.8 Why did you use weighted average F1-score instead of macro average F1-score for evaluation, and how does this affect the conclusions?

The weighted average F1-score weights class-specific F1-scores by their support (sample count). In our imbalanced dataset, the BENIGN class accounts for 86.87% of the unique frames, meaning the weighted F1-score is heavily dominated by the model's accuracy on benign data. This can mask poor performance on minority attack classes (e.g., if GAS throttle spoofing, which has only 25 samples, is missed, the weighted F1-score will still appear above 0.99). In contrast, a macro average F1-score weights all classes equally, which would penalize the model heavily for missing minority attacks. For safety-critical IDS evaluations, class-specific recall (Detection Rate) or macro average F1-score should be used as primary metrics to avoid false confidence.